CSA1515-Cloud computing and Big Data Analytics

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EXPERIMENT – 22

To write a procedure to run the virtual machine of different configuration and to check how many virtual machines can be utilized at a particular time.

Aim:

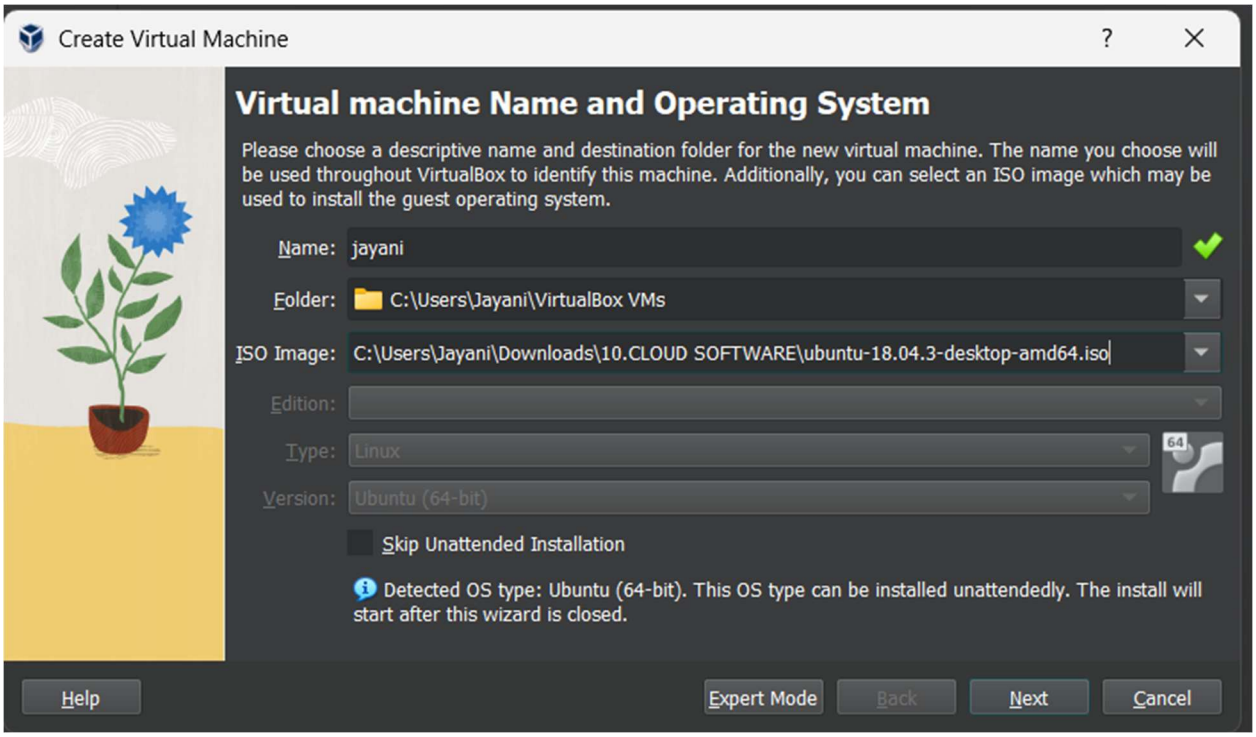
To write a procedure to run the virtual machine of different configuration and to check how many virtual machines can be utilized at a particular time.

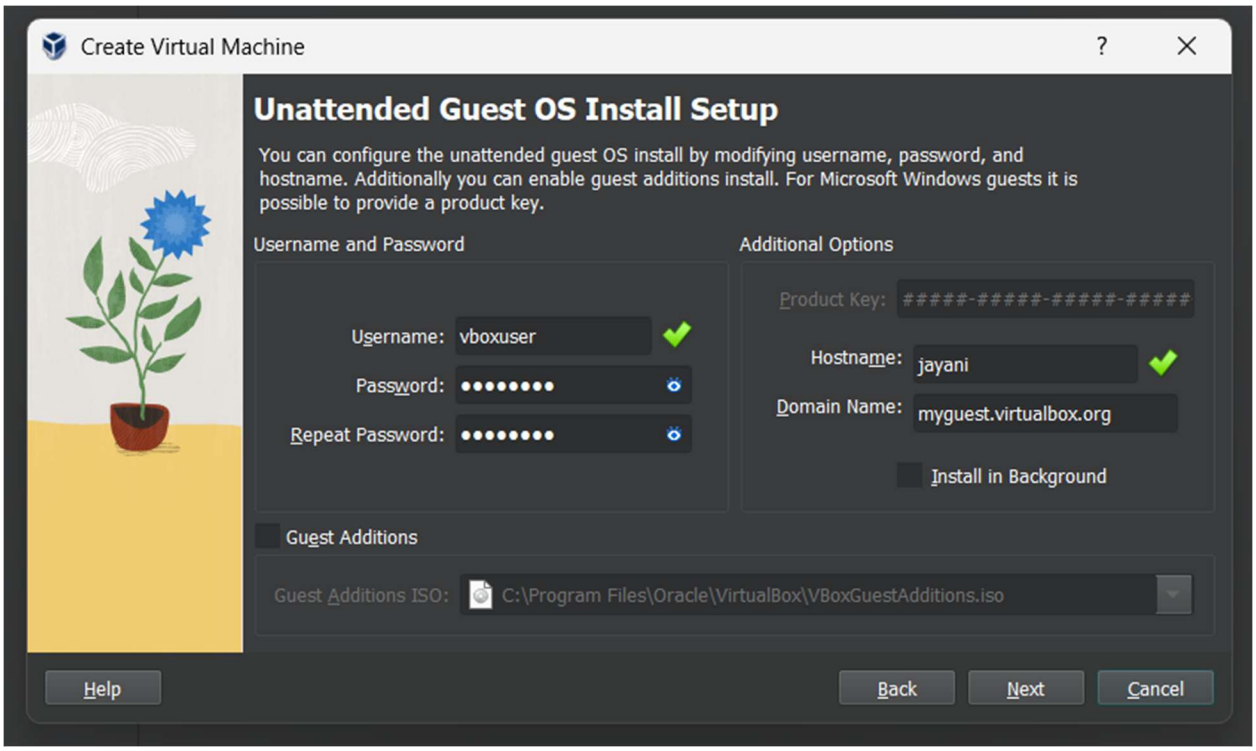
Short Procedure:

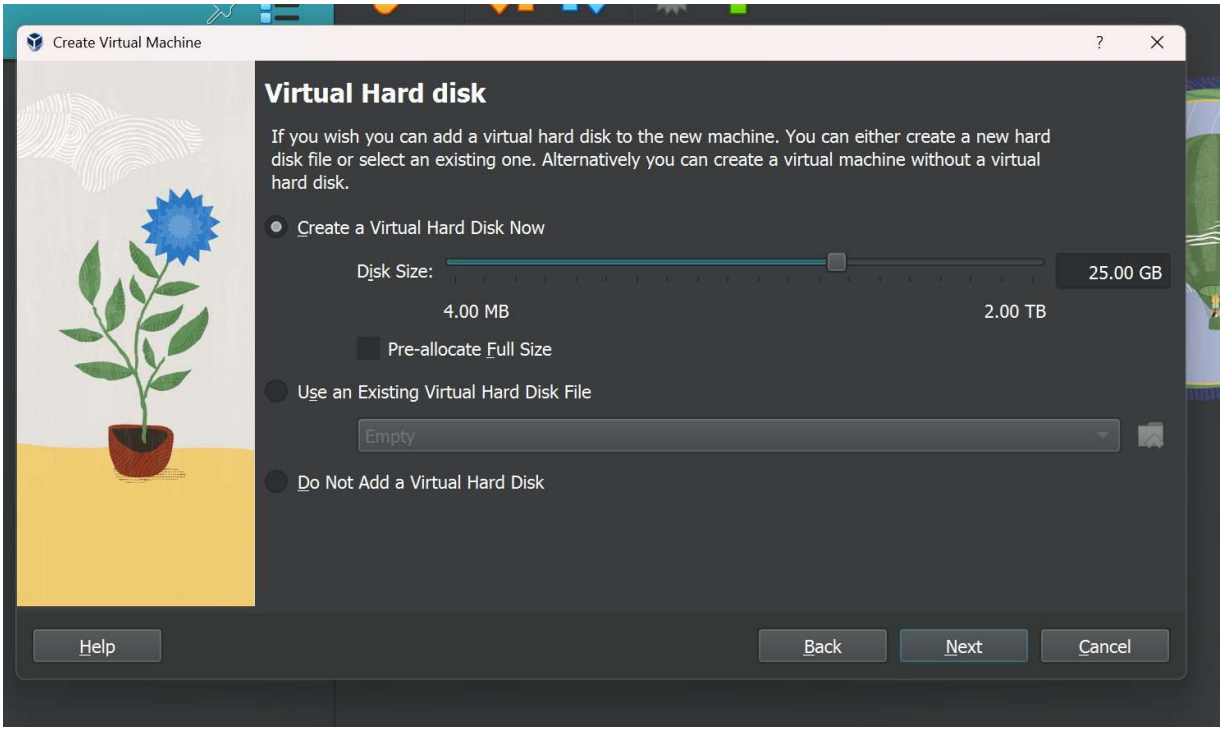
1. Open Oracle VirtualBox and create a new virtual machine.
2. Configure the VM with different RAM, CPU, and storage settings.
3. Install the required operating system in each virtual machine.
4. Start the virtual machines one by one and observe the system performance.
5. Continue running VMs until the available RAM/CPU resources are nearly utilized.
6. Record the number of VMs running simultaneously and their configurations.
7. Compare the configurations and determine the maximum number of VMs that can be utilized at a particular time.

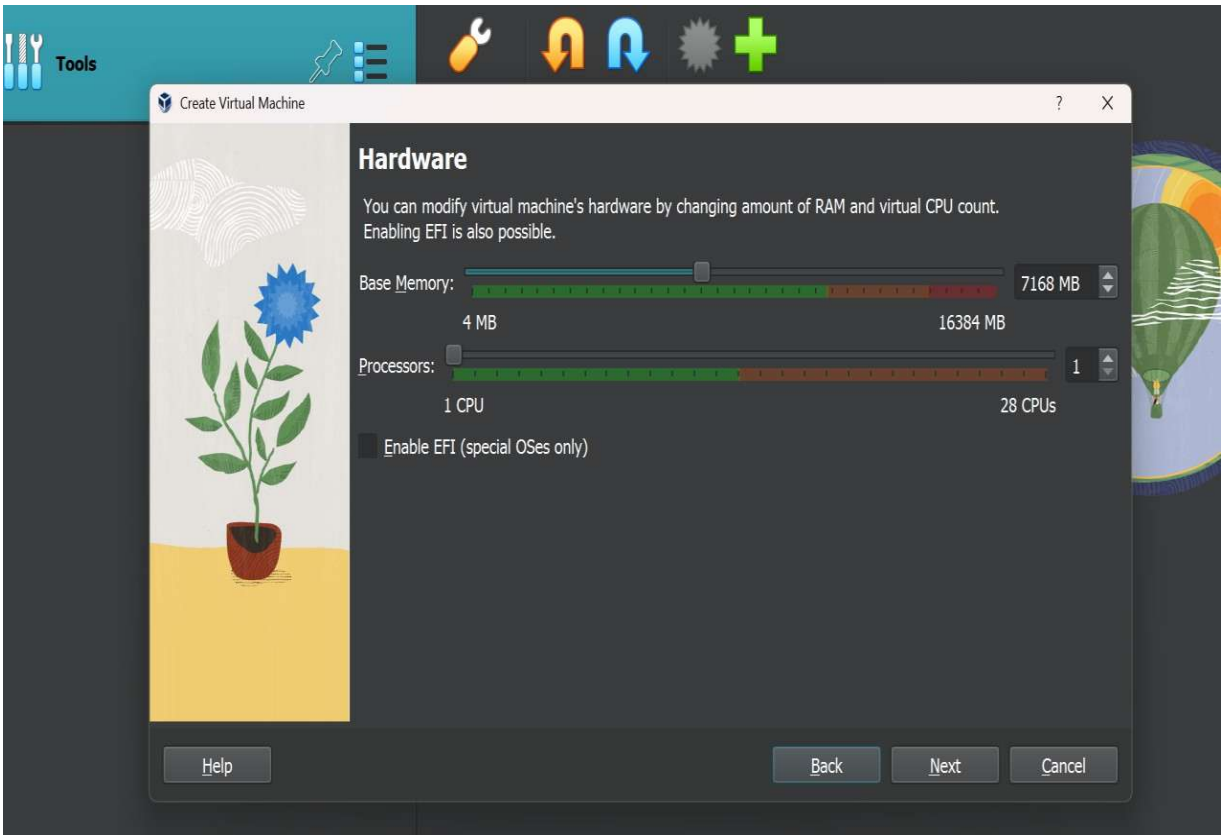
Result:

The number of virtual machines that can run simultaneously depends on the host system's available RAM, CPU, and storage resources.


Create Virtual Machine
Virtual machine Name and Operating System
Please choose a descriptive name and destination folder for the new virtual machine. The name you choose will be used throughout VirtualBox to identify this machine. Additionally, you can select an ISO image which may be used to install the guest operating system.
Name: jayani
Folder: C:\Users\Jayani\VirtualBox VMs
ISO Image: C:\Users\Jayani\Downloads\10.CLOUD SOFTWARE\ubuntu-18.04.3-desktop-amd64.iso
Edition:
Type: Linux
Version: Ubuntu (64-bit)
Skip Unattended Installation
Detected OS type: Ubuntu (64-bit). This OS type can be installed unattendedly. The install will start after this wizard is closed.
Help
Expert Mode
Back
Next
Cancel


Create Virtual Machine
Unattended Guest OS Install Setup
You can configure the unattended guest OS install by modifying username, password, and hostname. Additionally you can enable guest additions install. For Microsoft Windows guests it is possible to provide a product key.
Username and Password
Username: vboxuser
Password:
Repeat Password:
Additional Options
Product Key: #####-#####-#####-#####
Hostname: jayani
Domain Name: myguest.virtualbox.org
Install in Background
Guest Additions
Guest Additions ISO: C:\Program Files\Oracle\VirtualBox\VBoxGuestAdditions.iso
Help
Back
Next
Cancel


Create Virtual Machine
Virtual Hard disk
If you wish you can add a virtual hard disk to the new machine. You can either create a new hard disk file or select an existing one. Alternatively you can create a virtual machine without a virtual hard disk.
Create a Virtual Hard Disk Now
Disk Size:
25.00 GB
4.00 MB
2.00 TB
Pre-allocate Full Size
Use an Existing Virtual Hard Disk File
Empty
Do Not Add a Virtual Hard Disk
Help
Back
Next
Cancel


Tools
Create Virtual Machine
Hardware
You can modify virtual machine's hardware by changing amount of RAM and virtual CPU count.
Enabling EFI is also possible.
Base Memory:
7168 MB
4 MB
16384 MB
Processors:
1
1 CPU
28 CPUs
Enable EFI (special OSes only)
Help
Back
Next
Cancel

Record the number of VMs running simultaneously and their configurations.

VM Name	Operating System	RAM (MB)	CPU (Cores)	Storage (GB)	Status
Experiment1_VM	Ubuntu (64-bit)	2048	1	10.00	Running
Experiment2_VM	Ubuntu (64-bit)	2048	1	10.00	Running
Experiment3_VM	Ubuntu (64-bit)	10240	3	10.00	Running

VirtualBox Manager – Running Virtual Machines



Experiment1_VM (2 GB RAM, 1 CPU, 10 GB Storage)



Experiment2_VM (2 GB RAM, 1 CPU, 10 GB Storage)



Experiment3_VM (10 GB RAM, 3 CPU, 10 GB Storage)

Comparison and Maximum Number of VMs

Configuration	Total RAM Used (GB)	Total CPU Cores Used	Total Storage Used (GB)	No. of VMs Running	System Performance	Remarks
VM 1	2 GB	1	10 GB	1	Excellent	Smooth Performance
VM 1 + VM 2	4 GB	2	20 GB	2	Good	Slight Delay Observed
VM 1 + VM 2 + VM 3	14 GB	5	30 GB	3	Acceptable	High RAM Usage
Maximum VMs Running Simultaneously				3 VMs	14 GB RAM, 5 CPU Cores, 30 GB Storage	Beyond this system slows down

Conclusion: In this system configuration, a maximum of 3 virtual machines can be utilized simultaneously with acceptable performance.